

MODEL QUESTION PAPER

TED (10) 4049/ 5051

REVISION 2010

FIFTH SEMESTER EXAMINATION IN ENGINEERING / TECHNOLOGY

(Common to EL,EC,AE ,EA and EP)

AUDIO AND VIDEO SYSTEMS

(Maximum marks :100)

Time : 3 Hrs

PART A

I Answer the following questions in one or two sentences. Each question carries 2 marks .

1. Define intensity of sound in phon
2. Illustrate the need of Public Address System
3. Mention the significance of Kell factor
4. State the necessity of delayed AGC in Television receiver
5. What is compatibility and reverse compatibility in TV system

5X2 = 10 marks

PART B

II Answer any five of the following. Each Question carries 6 marks each

1. Sketch the constructional details of moving coil microphone. Write the principle of its working
2. What are the requirements of HI –FI system
3. Draw the standard composite video signals for two lines in CCIR system B
4. Enumerate the requirements for the choice of IF and local oscillator frequency of a T V receiver
5. Why (G-Y) signal is not used for colour TV transmission
6. Vestigial sideband transmission is used in TV system. Give its importance with necessary diagram
7. Compare NTSC and PAL system

5x6 = 30 marks

PART C

(Answer one full question from each module. Each question carries 15 marks)

MODULE I

III

- (a) Explain the principle of magnetic recording with neat diagram (8 marks)
- (b) Explain the concept of Dolby system (7 marks)

OR

IV

- (a) With the help of neat sketch explain the working of an electro dynamic type loud speaker (8 marks)
- (b) Differentiate between monophonic and stereo phonic sound system (7 marks)

MODULE II

V

- (a) write the demerits of sequential scanning (5 marks)
- (b) Compare positive and negative modulation with sketches (10 marks)

OR

VI

- (a) Explain the reason for using serrated vertical sync pulses with necessary sketches (10 marks)
- (b) Draw the constructional details of monochrome picture tube (5 marks)

MODULE III

VII

- (a) How V sync and H sync pulses are separated in TV receiver (5 marks)
- (b) Explain the push pull AFC circuit (10 marks)

OR

VIII

- (a) With the help of block diagram describe the VHF Tuner (8 marks)
- (b) Explain sound section of a TV receiver with block diagram (7 marks)

MODULE IV

- XI Draw the block diagram of PAL Colour TV transmitter. Describe its working (15 marks)
- OR

- XII (a) Explain additive colour mixing (7 marks)
- (b) Briefly explain the following terms (8 marks)
 - (i) hue (ii) saturation (iii) chrominance (iv) luminance
